Sahaj Bhadja

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Summary

AI/ML Engineer with 2+ years building **Python/C++ systems**, **PyTorch pipelines**, concurrent inference workflows, performance benchmarks, and production AI platforms focused on throughput, latency, and reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Analysis of Algorithms, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Rebuilt **asynchronous LLM inference workflows** using FastAPI workers, job states, failure isolation, and concurrent scheduling, processing 1K+ AI jobs while tripling throughput and preventing cascading production failures.
- Profiled and tuned PostgreSQL and SQL Server queries across 10K+ records, reducing latency 35% through **indexing, validation, query optimization, and reproducible CI/CD releases** for production AI workloads.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Integrated GPT-4o, Claude, and Qwen-VL behind **multimodal inference pipelines**, evaluating 10K+ daily submissions while tracking tokens, costs, fallbacks, and model consistency across demanding high-volume production AI workloads.
- Designed concurrent Python processing with ThreadPoolExecutor, OCR, timeouts, and secure ingestion, handling 1K+ files per batch while reducing turnaround 60% through **parallel execution and reliability tuning** reliably.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a **PyTorch Decision Transformer** for offline reinforcement learning, processing 100K+ trajectory logs through reproducible evaluations and improving long-horizon autonomous-driving planning performance 18% across controlled simulation experiments.
- Developed spatial-temporal encoding with HD maps, ego trajectories, multi-head attention, AdamW, and cosine scheduling, improving **closed-loop model performance** 15% across reproducible simulation baselines and research evaluations consistently.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Optimized distributed API and ETL components, improving response times 30% and pipeline efficiency 68% through **performance analysis, object-oriented design, AWS Glue, S3, and structured validation** workflows reliably.
- Automated builds and 20+ unit and integration suites with Jenkins, PyTest, Mockito, and virtual machines, strengthening **regression testing and performance tracking** across Windows, Linux, and macOS releases.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Designed eight object-oriented microservices for authentication, resources, billing, and notifications, synchronizing 2M+ records through **low-latency WebSockets and REST APIs** across distributed, real-time customer workflows and deployments reliably.
- Automated Docker packaging, Jenkins CI/CD, and test-driven releases, cutting deployment time 50% while using Sentry, Azure Monitor, and Kibana for **runtime monitoring, debugging, and failure diagnosis** daily.

Skills

Languages: Python, C++, SQL, Java, Data Structures, Algorithms, Numerical Computing, OOP
ML Systems: PyTorch, Transformers, LLM Inference, Reinforcement Learning, Quantization, Kernel Fusion, Graph Optimization, CUDA, Triton, vLLM
Performance: Profiling, Benchmarking, Throughput, Latency, Memory Optimization, Memory Hierarchy, Caching, Scheduling, Parallel Processing, Distributed Systems
Infrastructure: Linux, Docker, Kubernetes, AWS, Azure, S3, Jenkins, GitHub Actions, CI/CD, Git
Quality: PyTest, Mockito, Unit Testing, Integration Testing, Regression Testing, Debugging, Monitoring, Failure Isolation

Projects

Inference Profiler | *Python, PyTorch, vLLM, CUDA, Triton, Quantization*

- Designed an **LLM inference benchmarking harness** comparing prefill and decode latency, throughput, memory use, batch scheduling, quantization, and fused kernels across reproducible single-device and distributed execution configurations.

Graph Execution Lab | *C++, Python, PyTorch, CUDA, Operator Fusion, Scheduling*

- Built a **model-graph optimization sandbox** tracing tensor operations, identifying memory-bound bottlenecks, evaluating operator fusion and partitioning, and visualizing scheduling dependencies across compiler, runtime, and accelerator execution layers.

CPU-scheduling Simulator | *C++, React, Scheduling Algorithms, Performance Analysis* | Live

- Implemented **performance-oriented scheduling algorithms** including FCFS, SJF, SRT, Round Robin, priority, and multilevel queues, validating correctness and visualizing turnaround and waiting-time behavior across 1K+ simulated process combinations.